

COL 703/7203

07th Sept.

Claims

Let X be an MCS. Then

i) for all formulas ϕ , $\phi \in X$ iff $\neg\phi \notin X$

ii) for all formulas ϕ, ψ ,
 $\phi \vee \psi \in X$ iff $\phi \in X$
or $\psi \in X$.

i) $\{\phi, \neg\phi\}$ is inconsistent.

At least one of $\phi, \neg\phi$ must be there.

$X_j \cup \{\phi\}$ is inconsistent.

$X_k \cup \{\neg\phi\}$ is inconsistent.

$X_k \cup \{\phi\}$ is also inconsistent.

$\{\psi_{11}, \psi_{12}, \dots, \psi_{1n_1}, \phi\}$ is inconsistent

$\{\psi_{21}, \psi_{22}, \dots, \psi_{2n_2}, \neg\phi\}$ is also inconsistent.

$\{\psi_{11}, \dots, \psi_{1n_1}, \psi_{21}, \dots, \psi_{2n_2}\} \cup \{\phi\}$
 $\cup \{\neg\phi\}$

are both
inconsistent -
why? Quiz 2 (Q1/Q2).

$$\psi_1, \dots, \psi_m$$

$$\psi = \bigwedge_{i=1}^m \psi_i$$

$\psi \wedge \phi$ is inconsistent

$\psi \wedge \neg \phi$ is inconsistent.

$$\vdash \neg(\psi \wedge \phi)$$

$$\vdash \neg(\psi \wedge \neg \phi)$$

$$\vdash \neg \neg(\neg \psi \vee \neg \phi)$$

$$\vdash \neg \neg(\neg \psi \vee \neg \phi)$$

$$\psi \rightarrow \phi$$

$$\psi \rightarrow \neg \phi$$



if they are both derivable
then $\neg \psi$ is also
derivable.

$$(\psi \rightarrow \neg \phi) \rightarrow ((\psi \rightarrow \phi) \rightarrow \neg \psi)$$

$\therefore \psi$ must not be
consistent.

ii) Suppose $\phi \vee \psi \in X$ but

$$\phi \notin X$$

$$\psi \notin X$$

From part (i) we know that

$$\begin{cases} \neg \phi \in X \\ \neg \psi \in X \end{cases}$$

Then, $\{\neg \phi, \neg \psi, \phi \vee \psi\}$ is inconsistent. } Exercise.

Exercise: prove the other direction as well.

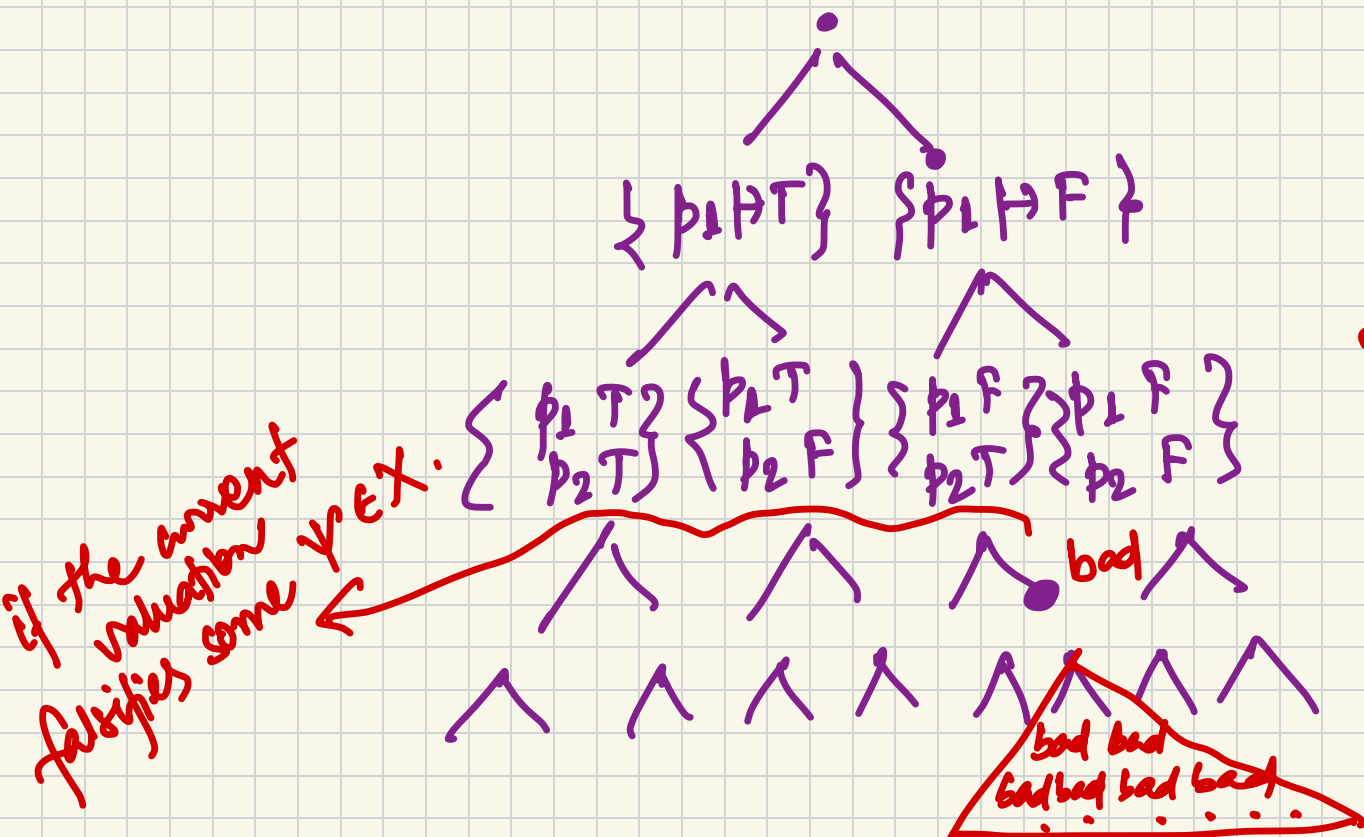
Compactness Theorem / Finite Satisfiability

Lemma

Let $X \subseteq \mathcal{F}$. Then X
is satisfiable iff every
 $Y \subseteq_{\text{fin}} X$ is satisfiable.

One direction is trivial.

Let $\{p_1, p_2, \dots\}$ be the propositional atoms used in Φ .



$\top$ is the tree after pruning all "bad" nodes that also have a "bad" ancestor.

Claim : If X is not satisfiable then T (the pruned tree) has only finitely many nodes.

Why? Because if it has infinitely many nodes then X is actually satisfiable.

Why? Because if it has infinitely many nodes then (because it is finitely branching), it has an infinite path.

König's lemma.

In other words,
this valuation
satisfies $\vdash$.

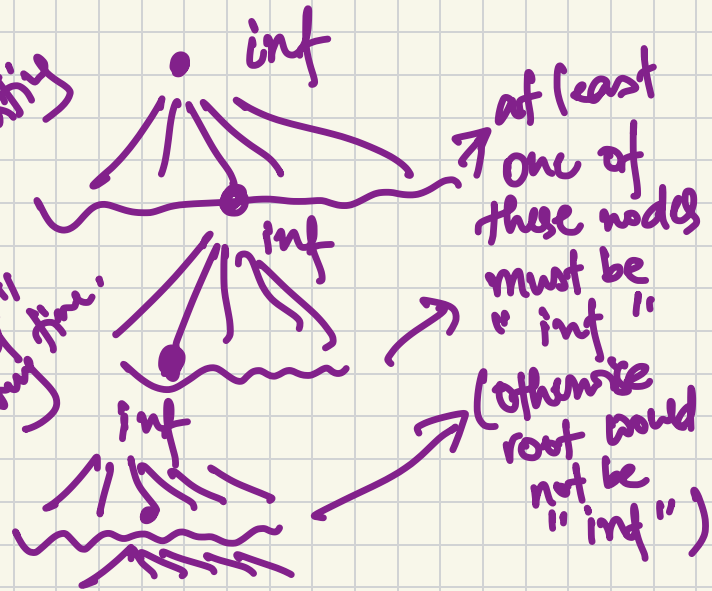
That infinite path corresponds to a valuation which does not falsify any $\psi \in X$ (because otherwise the path would have a bad node which would be the last node on that path).

König's Lemma

Let T be a finitely branching tree. If T has infinitely many nodes, then it has an infinite path.

Call a node "inf" if the subtree rooted at it has infinitely many nodes. Clearly, root is "inf".

An infinite path can be obtained by starting at the root and picking an "inf" child every time.



Now because the pruned tree has only finitely many nodes, it will have only finitely many leaves.

Each leaf is a "bad" node, so there is some φ_i corresponding to it which is falsified by the valuation corresponding to that leaf.

It is easy to see that the set of all such φ_i 's is a finite set which is not satisfiable.